

Design procedure for low-pass digital Butterworth IIR filter

The low-pass digital Butterworth filter is designed as per the following steps:

- Let A_1 = Gain at a passband frequency ω_1
 A_2 = Gain at a stopband frequency ω_2
 Ω_1 = Analog frequency corresponding to ω_1
 Ω_2 = Analog frequency corresponding to ω_2

Step 1 Choose the type of transformation, i.e., either bilinear or impulse invariant transformation

Step 2 Calculate the ratio of analog edge frequencies Ω_2/Ω_1 .

For bilinear transformation

$$\Omega_1 = \frac{2}{T} \tan \frac{\omega_1}{2}, \Omega_2 = \frac{2}{T} \tan \frac{\omega_2}{2} \quad \therefore \frac{\Omega_2}{\Omega_1} = \frac{\tan \omega_2/2}{\tan \omega_1/2}$$

For impulse invariant transformation,

$$\Omega_1 = \frac{\omega_1}{T}, \Omega_2 = \frac{\omega_2}{T} \quad \therefore \frac{\Omega_2}{\Omega_1} = \frac{\omega_2}{\omega_1}$$

Step 3 Decide the order N of the filter. The order N should be such that

$$N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] / \left[\frac{1}{A_1^2} - 1 \right] \right\}}{\log \frac{\Omega_2}{\Omega_1}}$$

Choose N such that it is an integer just greater than or equal to the value obtained above

If ϕ is the angle of a valid pole w.r.t. the X-axis, then the pole and its conjugate are located at $[\omega_c(\cos \phi \pm j \sin \phi)]$.

Properties of Butterworth filters

1. The Butterworth filters are all pole designs (i.e. the zeros of the filters exist at ∞).
2. The filter order N completely specifies the filter.

Step 4 Calculate the analog cutoff frequency $\Omega_c = \frac{\Omega_1}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}}$

For bilinear transformation $\Omega_c = \frac{\frac{2}{T} \tan \omega_1/2}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}}$

~~For impulse invariant transformation $\Omega_c = \frac{\omega_1/T}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}}$~~

Step 5 Determine the transfer function of the analog filter. Let $H_a(s)$ be the transfer function of the analog filter. When the order N is even, for unity dc gain filter, $H_a(s)$ is given by

$$H_a(s) = \prod_{k=1}^{N/2} \frac{\Omega_c^2}{s^2 + b_k \Omega_c s + \Omega_c^2}$$

When the order N is odd, for unity dc gain filter, $H_a(s)$ is given by

$$H_a(s) = \frac{\Omega_c}{s + \Omega_c} \prod_{k=1}^{N-1} \frac{\Omega_c^2}{s^2 + b_k \Omega_c s + \Omega_c^2}$$

The coefficient b_k is given by

$$b_k = 2 \sin \left[\frac{(2k-1)\pi}{2N} \right]$$

For normalized case, $\Omega_c = 1$ rad/s

Step 6 Using the chosen transformation, transform the analog filter transfer function $H_a(s)$ to digital filter transfer function $H(z)$.

Step 7 Realize the digital filter transfer function $H(z)$ by a suitable structure.

3. The magnitude response approaches the ideal response as the value of N increases.
4. The magnitude is maximally flat at the origin.
5. The magnitude is monotonically decreasing function of Ω .
6. At the cutoff frequency Ω_c , the magnitude of normalized Butterworth filter is $1/\sqrt{2}$. Hence the dB magnitude at the cutoff frequency will be 3 dB less than the maximum value.

EXAMPLE 8.17 Design a Butterworth digital filter using the bilinear transformation. The specifications of the desired low-pass filter are:

$$0.9 \leq |H(\omega)| \leq 1; \quad 0 \leq \omega \leq \frac{\pi}{2}$$

$$|H(\omega)| \leq 0.2; \quad \frac{3\pi}{4} \leq \omega \leq \pi$$

with $T = 1$ s

Solution: The Butterworth digital filter is designed as per the following steps.
From the given specification, we have

$$A_1 = 0.9 \text{ and } \omega_1 = \frac{\pi}{2}$$

$$A_2 = 0.2 \text{ and } \omega_2 = \frac{3\pi}{4} \quad \text{and } T = 1 \text{ s}$$

Step 1 Choice of the type of transformation

Here the bilinear transformation is already specified.

Step 2 Determination of the ratio of the analog filter's edge frequencies, Ω_2/Ω_1

$$\Omega_2 = \frac{2}{T} \tan \frac{\omega_2}{2} = \frac{2}{1} \tan \left[\frac{(3\pi/4)}{2} \right] = 2 \tan \frac{3\pi}{8} = 4.828$$

$$\Omega_1 = \frac{2}{T} \tan \frac{\omega_1}{2} = \frac{2}{1} \tan \left[\frac{(\pi/2)}{2} \right] = 2 \tan \frac{\pi}{4} = 2$$

$$\therefore \frac{\Omega_2}{\Omega_1} = \frac{4.828}{2} = 2.414$$

Step 3 Determination of the order of the filter N

$$N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] \left/ \left[\frac{1}{A_1^2} - 1 \right] \right\}}{\log \frac{\Omega_2}{\Omega_1}}$$

$$\begin{aligned}
 & \geq \frac{1}{2} \frac{\log \left\{ \frac{1}{(0.2)^2} - 1 \right\} / \left[\frac{1}{(0.9)^2} - 1 \right]}{\log 1.207} \\
 & \geq \frac{1}{2} \frac{\log \{24/0.2345\}}{\log 2.414} \geq 2.626
 \end{aligned}$$

Since $N \geq 2.626$, choose $N = 3$.

Step 4 Determination of the analog cutoff frequency Ω_c (i.e., -3 dB frequency)

$$\Omega_c = \frac{\Omega_1}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}} = \frac{2}{\left[\frac{1}{0.9^2} - 1 \right]^{1/2 \times 3}} = 2.5467$$

Step 5 Determination of the transfer function of the analog Butterworth filter $H_a(s)$

For odd N , we have $H_a(s) = \frac{\Omega_c}{s + \Omega_c} \prod_{k=1}^{\frac{N-1}{2}} \frac{\Omega_c^2}{s^2 + b_k \Omega_c s + \Omega_c^2}$

where $b_k = 2 \sin \left[\frac{(2k-1)\pi}{2N} \right]$

For $N = 3$, we have

$$H_a(s) = \frac{\Omega_c}{s + \Omega_c} \frac{\Omega_c^2}{s^2 + b_1 \Omega_c s + \Omega_c^2}$$

where $b_1 = 2 \sin \left[\frac{(2 \times 1 - 1)\pi}{2 \times 3} \right] = 2 \sin \frac{\pi}{6} = 1$

Therefore, $H_a(s) = \left(\frac{2.5467}{s + 2.5467} \right) \left(\frac{(2.5467)^2}{s^2 + 1(2.5467)s + (2.5467)^2} \right)$

Step 6 Conversion of $H_a(s)$ into $H(z)$

Since bilinear transformation is to be used, the digital filter transfer function is:

$$H(z) = H_a(s) \Big|_{s=\frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)} = H_a(s) \Big|_{s=2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

$$H(z) = \left(\frac{2.5467}{2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 2.5467} \right) \left(\frac{(2.5467)^2}{\left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right]^2 + 2.5467 \left[2 \frac{1-z^{-1}}{1+z^{-1}} \right] + (2.5467)^2} \right)$$

$$= \frac{0.2332(1+z^{-1})^3}{1 + 0.4394z^{-1} + 0.3845z^{-2} + 0.0416z^{-3}}$$

EXAMPLE 8.19 Design a low-pass Butterworth digital filter to give response of 3 dB or less for frequencies upto 2 kHz and an attenuation of 20 dB or more beyond 4 kHz. Use the bilinear transformation technique and obtain $H(z)$ of the desired filter.

Solution: The specifications of the desired filter are given in terms of dB attenuation and frequency in Hz. First the gain is to be expressed as a numerical value and frequency in rad/s.

Here attenuation at passband frequency (ω_1) = 3 dB

Therefore, gain at passband edge frequency (ω_1) is $k_1 = -3$ dB

$$\therefore A_1 = 10^{k_1/20} = 10^{-3/20} = 0.707 = \frac{1}{\sqrt{2}}$$

Attenuation at stopband frequency (ω_2) = 20 dB

Therefore, gain at stopband edge frequency (ω_2) is $k_2 = -20$ dB

$$\therefore A_2 = 10^{k_2/20} = 10^{-20/20} = 0.1$$

Passband edge frequency = 2 kHz,

Stopband edge frequency = 4 kHz,

The design is performed as given below.

Let the sampling frequency be 10000 Hz.

$$\therefore \text{Normalized } \omega_1 = 2\pi \frac{f_1}{f_s} = 2\pi \frac{2000}{10000} = 0.4\pi$$

$$\text{Normalized } \omega_2 = 2\pi \frac{f_2}{f_s} = 2\pi \frac{4000}{10000} = 0.8\pi$$

Step 1 Bilinear transformation is chosen

Step 2 Ratio of analog filter edge frequencies Ω_2/Ω_1

$$\Omega_1 = \frac{2}{T} \tan \frac{\omega_1}{2} = \frac{2}{T} \tan \frac{0.4\pi}{2} = 14530.8 \text{ rad/s}$$

$$\Omega_2 = \frac{2}{T} \tan \frac{\omega_2}{2} = \frac{2}{T} \tan \frac{0.8\pi}{2} = 61553.6 \text{ rad/s}$$

$$\frac{\Omega_2}{\Omega_1} = \frac{\tan \frac{\omega_2}{2}}{\tan \frac{\omega_1}{2}} = \frac{\tan 0.4\pi}{\tan 0.2\pi} = 4.236$$

Step 3 Order of the filter

$$N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] / \left[\frac{1}{A_1^2} - 1 \right] \right\}}{\log \frac{\Omega_2}{\Omega_1}}$$

$$\geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{(0.1)^2} - 1 \right] / \left[\frac{1}{(1/\sqrt{2})^2} - 1 \right] \right\}}{\log 4.236}$$

$$\geq \frac{1}{2} \frac{\log [99/1]}{\log 4.236} \geq 1.59$$

$$N = 2$$

Step 4 Analog cutoff frequency Ω_c

$$\Omega_c = \frac{\Omega_1}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}} = \frac{1.4530}{\left[\frac{1}{(1/\sqrt{2})^2} - 1 \right]^{1/2 \times 2}} = 1.4530$$

$$\text{Unnormalized } \Omega_c = f_s \times 1.4530 = 14530 \text{ rad/s}$$

Step 5 Transfer function $H_a(s)$

$$\text{For } N = 2, H_a(s) = \frac{\Omega_c^2}{s^2 + b_1 \Omega_c s + \Omega_c^2}$$

$$\text{where } b_1 = 2 \sin \left[\frac{(2 \times 1 - 1)\pi}{2 \times 2} \right] = 2 \sin \frac{\pi}{4} = 1.414$$

$$H_a(s) = \frac{(14530)^2}{s^2 + 1.414 \times 14530 s + (14530)^2}$$

$$= \frac{2.1112 \times 10^8}{s^2 + 20545.42 s + 2.1112 \times 10^8}$$

Step 6 Conversion of $H_a(s)$ into $H(z)$

$$H(z) = H_a(s) \Big|_{s=\frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)} = H_a(s) \Big|_{s=20000 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

$$H(z) = \frac{2.112 \times 10^8}{\left[20 \times 10^3 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right]^2 + 2.0545 \times 10^4 \times 20 \times 10^3 \left[\frac{1-z^{-1}}{1+z^{-1}} \right] + 2.112 \times 10^8}$$

$$= \frac{0.528}{2.5552 - 0.946z^{-1} + 0.5008z^{-2}}$$

EXAMPLE 8.20 Design a low-pass Butterworth filter using the bilinear transformation method for satisfying the following constraints:

Passband: 0–400 Hz

Passband ripple: 2 dB

Sampling frequency: 10 kHz

Stopband: 2.1–4 kHz

Stopband attenuation: 20 dB

Solution: Given

$$\alpha_1 = 2 \text{ dB}, \quad \therefore k_1 = -2 \text{ dB} \quad \text{and} \quad A_1 = 10^{k_1/20} = 10^{-2/20} = 0.794$$

$$\alpha_2 = 20 \text{ dB}, \quad \therefore k_2 = -20 \text{ dB} \quad \text{and} \quad A_2 = 10^{k_2/20} = 10^{-20/20} = 0.1$$

Step 1 Type of transformation

Bilinear transformation is already specified.

Step 2 Ratio of analog edge frequencies Ω_2/Ω_1 .

Here $f_s = 10 \text{ kHz}$

Passband edge frequency $f_1 = 400 \text{ Hz}$

Stopband edge frequency $f_2 = 2.1 \text{ kHz}$

Normalizing the frequencies, we have

$$\omega_1 = 2\pi \frac{f_1}{f_s} = 2\pi \times \frac{400}{10000} = 0.25 \text{ rad}$$

$$\omega_2 = 2\pi \frac{f_2}{f_s} = 2\pi \times \frac{2100}{10000} = 1.319 \text{ rad}$$

Therefore, the analog filter edge frequencies are:

$$\Omega_1 = \frac{2}{T} \tan \frac{\omega_1}{2} = 2 \times 10000 \tan \frac{0.25}{2} = 2513.102 \text{ rad/s}$$

and $\Omega_2 = \frac{2}{T} \tan \frac{\omega_2}{2} = 2 \times 10000 \tan \frac{1.319}{2} = 15,506.08 \text{ rad/s}$

$$\frac{\Omega_2}{\Omega_1} = \frac{15506.08}{2513.102} = 6.1703$$

Step 3 Order of the filter N

$$N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] / \left[\frac{1}{A_1^2} - 1 \right] \right\}}{\log \left(\frac{\Omega_2}{\Omega_1} \right)} \quad \text{or} \quad N \geq \frac{1}{2} \frac{\log \left\{ \frac{10^{0.1 A_2 \text{ dB}} - 1}{10^{0.1 A_1 \text{ dB}} - 1} \right\}}{\log (6.1703)}$$

$$\text{i.e. } N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{(0.1)^2} - 1 \right] / \left[\frac{1}{(0.794)^2} - 1 \right] \right\}}{\log (6.1703)} \quad \text{or} \quad N \geq \frac{1}{2} \frac{\log \left\{ \frac{10^{0.1 \times 20 \text{ dB}} - 1}{10^{0.1 \times 2 \text{ dB}} - 1} \right\}}{\log (6.1703)}$$

$$\text{i.e. } N \geq 1.409 \approx 2 \quad \text{or} \quad N \geq 1.410 \approx 2$$

Step 4 The cutoff frequency Ω_c

$$\Omega_c = \frac{\Omega_2}{\left[\frac{1}{A_2^2} - 1 \right]^{1/2N}} \quad \text{or} \quad \Omega_c = \frac{\Omega_2}{[10^{0.1 A_2 \text{ dB}} - 1]^{1/2N}}$$

i.e.

$$\Omega_c = \frac{15506.08}{\left[\frac{1}{(0.1)^2} - 1 \right]^{1/2 \times 2}} = 4915.7 \quad \text{or} \quad \Omega_c = \frac{15506.08}{[10^{0.1 \times 20} - 1]^{1/2 \times 2}} = 4915.788 \text{ rad/s}$$

Step 5 The system function $H_a(s)$

$$\begin{aligned} H_a(s) &= \frac{\Omega_c^2}{s^2 + b_1 \Omega_c s + \Omega_c^2} \quad \text{where } b_1 = 2 \sin \frac{(2 \times 1 - 1)\pi}{2 \times 2} = 1.414 \\ &= \frac{(4915.788)^2}{s^2 + 1.414 \times 4915.788 s + (4915.788)^2} \\ &= \frac{2.416 \times 10^7}{s^2 + 6950.92 s + 2.416 \times 10^7} \end{aligned}$$

Step 6 Digital transfer function $H(z)$

$$H(z) = H_a(s) \bigg|_{s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

$$= \left[\frac{2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right]^2 + 6950.92 \times \frac{2 \left[\frac{1-z^{-1}}{1+z^{-1}} \right]}{T} + 2.416 \times 10^7$$

$$= \left[20000 \times \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right]^2 + 6950.92 \times 20000 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 2.416 \times 10^7$$

$$= \frac{0.042 + 0.085z^{-1} + 0.042z^{-2}}{1 - 1.335z^{-1} + 0.506z^{-2}}$$

The poles are given by $P_k = \pm (\Omega_c) \left[e^{j \frac{(2k+N+1)\pi}{2N}} \right], k = 0, 1 < N$

$$P_0 = \pm (\Omega_c) e^{j \left(\frac{3\pi}{4} \right)} = 4.915.788(-0.707 + j0.707) = -3475.6 + j3475.46$$

$$P_1 = \pm (\Omega_c) e^{j \left(\frac{5\pi}{4} \right)} = -3475.6 - j3475.6$$

EXAMPLE 8.21 A digital low-pass filter is required to meet the following specifications.

Passband attenuation ≤ 1 dB

Passband edge = 4 kHz

Stopband attenuation ≥ 40 dB

Stopband edge = 8 kHz

Sampling rate = 24 kHz

The filter is to be designed by performing the bilinear transformation on an analog system function. Design the Butterworth filter.

Solution: Given $\alpha_1 = 1$ dB, $\therefore k_1 = -1$ dB and $A_1 = 10^{k_1/20} = 10^{-1/20} = 0.8912$

$\alpha_2 = 40$ dB, $\therefore k_2 = -40$ dB and $A_2 = 10^{k_2/20} = 10^{-40/20} = 0.01$

Since $f_s = 24$ kHz, normalized angular frequencies are:

$$f_1 = 4 \text{ kHz}, \therefore \omega_1 = \frac{2\pi f_1}{f_s} = 2\pi \times \frac{4000}{24000} = 1.047 \text{ rad/s}$$

$$f_2 = 8 \text{ kHz}, \therefore \omega_2 = \frac{2\pi f_2}{f_s} = 2\pi \times \frac{8000}{24000} = 2.094 \text{ rad/s}$$

The Butterworth filter is designed as follows:

Step 1 Type of transformation

Bilinear transformation is already specified.

Step 2 Ratio of analog edge frequencies, Ω_2/Ω_1

$$\Omega_1 = \frac{2}{T} \tan \frac{\omega_1}{2} = 2 \times 24000 \tan \frac{1.047}{2} = 27706.49 \text{ rad/s}$$

$$\Omega_2 = \frac{2}{T} \tan \frac{\omega_2}{2} = 2 \times 24000 \tan \frac{2.094}{2} = 83100.52 \text{ rad/s}$$

$$\therefore \frac{\Omega_2}{\Omega_1} = \frac{83000.52}{27706.49} = 2.9957$$

Step 3 Order of the filter N

$$N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] / \left[\frac{1}{A_1^2} - 1 \right] \right\}}{\log (\Omega_2 / \Omega_1)}$$

$$\text{or } N \geq \frac{1}{2} \frac{\log \left(\frac{10^{0.1 A_2 \text{ dB}} - 1}{10^{0.1 A_1 \text{ dB}} - 1} \right)}{\log (\Omega_2 / \Omega_1)}$$

$$\geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{(0.01)^2} - 1 \right] / \left[\frac{1}{(0.8912)^2} - 1 \right] \right\}}{\log (2.9957)}$$

$$\text{or } N \geq \frac{1}{2} \frac{\log \left(\frac{10^{0.1 \times 40} - 1}{10^{0.1 \times 1} - 1} \right)}{\log (2.9957)}$$

$$\geq \frac{1}{2} \frac{\log \{9999/0.2590\}}{\log (2.9957)}$$

$$\text{or } N \geq \frac{1}{2} \frac{\log (9999/0.2589)}{\log (2.9957)}$$

$$\geq \frac{1}{2} \frac{4.586}{0.476}$$

$$\text{or } N \geq \frac{1}{2} \frac{4.586}{0.476}$$

$$\geq 4.8 \approx 5$$

$$\text{or } N \geq 4.8 \approx 5$$

Step 4 The cutoff frequency Ω_c

$$\Omega_c = \frac{\Omega_1}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}}$$

$$\text{or } \Omega_c = \frac{\Omega_1}{[10^{0.1 A_1 \text{ dB}} - 1]^{1/2N}}$$

$$= \frac{27706.49}{\left[\frac{1}{(0.8912)^2} - 1 \right]^{1/2 \times 5}}$$

$$\text{or } \Omega_c = \frac{27706.49}{[10^{0.1 \times 1} - 1]^{1/2 \times 5}}$$

$$= 31,715 \text{ rad/s}$$

$$\text{or } \Omega_c = 31,715 \text{ rad/s}$$

Step 5 Analog filter transfer function $H_a(s)$

$$\text{For } N = 5, H_a(s) = \left(\frac{\Omega_c}{s + \Omega_c} \right) \left(\frac{\Omega_c^2}{s^2 + b_1 \Omega_c s + \Omega_c^2} \right) \left(\frac{\Omega_c^2}{s^2 + b_2 \Omega_c s + \Omega_c^2} \right)$$

where

$$b_1 = 2 \sin \left(\frac{(2 \times 1 - 1) \pi}{2N} \right) = 2 \sin \frac{\pi}{10} = 0.618$$

$$b_2 = 2 \sin \left(\frac{(2 \times 2 - 1) \pi}{10} \right) = 2 \sin \frac{3\pi}{10} = 1.618$$

$$\therefore H_a(s) = \left(\frac{31708}{s + 31708} \right) \left(\frac{(31708)^2}{s^2 + 0.618 \times 31708 s + (31708)^2} \right) \\ \left(\frac{(31708)^2}{s^2 + 1.618 \times 31708 s + (31708)^2} \right)$$

Step 6 Digital filter function $H(z)$
Using the bilinear transformation, we have

$$H(z) = H_a(s) \Big|_{s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)} = H_a(s) \Big|_{s = 48000 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

$$= \left[\left(\frac{31708}{s + 31708} \right) \left(\frac{(31708)^2}{s^2 + 0.618 \times 31708 s + (31708)^2} \right) \right. \\ \left. \left(\frac{(31708)^2}{s^2 + 1.618 \times 31708 s + (31708)^2} \right) \right] \Big|_{s = 48000 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

EXAMPLE 8.22 Design a digital IIR low-pass filter with passband edge at 1000 Hz and stopband edge at 1500 Hz for a sampling frequency of 5000 Hz. The filter is to have passband ripple of 0.5 dB and a stopband ripple below 30 dB. Design a Butterworth filter using the bilinear transformation.

Solution: Given $f_s = 5000$ Hz, the normalized frequencies are given as:

$$\text{Passband edge } f_1 = 1000 \text{ Hz, } \therefore \omega_1 = 2\pi \frac{f_1}{f_s} = 2\pi \times \frac{1000}{5000} = 0.4\pi \text{ rad/s}$$

$$\text{Stopband edge } f_2 = 1500 \text{ Hz, } \therefore \omega_2 = 2\pi \frac{f_2}{f_s} = 2\pi \times \frac{1500}{5000} = 0.6\pi \text{ rad/s}$$

$$\alpha_1 = 0.5 \text{ dB, } \therefore k_1 = -0.5 \text{ dB and } A_1 = 10^{k_1/20} = 10^{-0.5/20} = 0.9446$$

$$\alpha_2 = 30 \text{ dB, } \therefore k_2 = -30 \text{ dB and } A_2 = 10^{k_2/20} = 10^{-30/20} = 0.0316$$

The Butterworth filter is designed as follows:

Step 1 Type of transformation.
Bilinear transformation is to be used.

Step 2 Ratio of analog filter edge frequencies, Ω_2/Ω_1

$$\Omega_1 = \frac{2}{T} \tan \frac{\omega_1}{2} = 2 \times 5000 \tan \frac{0.4\pi}{2} = 7265.425 \text{ rad/s}$$

$$\Omega_2 = \frac{2}{T} \tan \frac{\omega_2}{2} = 2 \times 5000 \tan \frac{0.6\pi}{2} = 13763.819 \text{ rad/s}$$

$$\frac{\Omega_2}{\Omega_1} = \frac{13763.819}{7265.425} = 1.8944$$

Step 3 Order of the filter N

$$N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] \middle/ \left[\frac{1}{A_1^2} - 1 \right] \right\}}{\log (\Omega_2/\Omega_1)}$$

$$\geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{(0.0316)^2} - 1 \right] \middle/ \left[\frac{1}{(0.9446)^2} - 1 \right] \right\}}{\log (1.844)}$$

$$\geq \frac{1}{2} \frac{\log \{1000.44/0.1207\}}{\log (1.844)}$$

$$\geq 7.35 \approx 8$$

Step 4 The cutoff frequency Ω_c

$$\Omega_c = \frac{\Omega_1}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}} = \frac{7265.425}{\left[\frac{1}{0.9446^2} - 1 \right]^{1/2 \times 8}} = 8292 \text{ rad/s}$$

Step 5 The system function $H_a(s)$

$$H_a(s) = \prod_{k=1}^{N/2} \frac{\Omega_c^2}{s^2 + b_k \Omega_c s + \Omega_c^2}$$

$$\text{where } b_k = 2 \sin \left[\frac{(2k-1)\pi}{2N} \right]$$

$$b_1 = 2 \sin \left[\frac{\pi}{16} \right] = 0.390$$

$$b_2 = 2 \sin \left[\frac{3\pi}{16} \right] = 1.111$$

$$b_3 = 2 \sin \left[\frac{5\pi}{16} \right] = 1.662$$

$$b_4 = 2 \sin \left[\frac{7\pi}{16} \right] = 1.961$$

$$H_a(s) = \left(\frac{(8292)^2}{s^2 + 0.39 \times 8292s + (8292)^2} \right) \left(\frac{(8292)^2}{s^2 + 1.111 \times 8292s + (8292)^2} \right) \\ \left(\frac{(8292)^2}{s^2 + 1.662 \times 8292s + (8292)^2} \right) \left(\frac{(8292)^2}{s^2 + 1.961 \times 8292s + (8292)^2} \right)$$

Step 6 Digital filter function $H(z)$
Using the bilinear transformation, we have

$$H(z) = H_a(s) \Big|_{s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)} = H_a(s) \Big|_{s = 10000 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

$$H(z) = \left[\left(\frac{(8292)^2}{s^2 + 3233.8s + (8292)^2} \right) \left(\frac{(8292)^2}{s^2 + 9212.4s + (8292)^2} \right) \right] \left[\left(\frac{(8292)^2}{s^2 + 13781.3s + (8292)^2} \right) \left(\frac{(8292)^2}{s^2 + 16260.6s + (8292)^2} \right) \right] \Big|_{s = 10000 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

EXAMPLE 8.23 Find the filter order for the following specifications:

$$\sqrt{0.5} \leq |H(\omega)| \leq 1 \quad 0 \leq \omega \leq \pi/2$$

$$|H(\omega)| \leq 0.2 \quad 3\pi/4 \leq \omega \leq \pi$$

with $T = 1$ s. Use the impulse invariant method.

Solution: Given $A_1 = \sqrt{0.5}$, $\omega_1 = \frac{\pi}{2}$

$$A_2 = 0.2, \quad \omega_2 = \frac{3\pi}{4}$$

$$T = 1 \text{ s.}$$

The impulse invariant transformation is to be used.

$$\frac{\Omega_2}{\Omega_1} = \frac{\omega_2/T}{\omega_1/T} = \frac{\omega_2}{\omega_1} = \frac{3\pi/4}{\pi/2} = 1.5$$

Order of the low-pass Butterworth filter N

$$\begin{aligned} N &\geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] / \left[\frac{1}{A_1^2} - 1 \right] \right\}}{\log (\Omega_2 / \Omega_1)} \\ &\geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{(0.2)^2} - 1 \right] / \left[\frac{1}{(\sqrt{0.5})^2} - 1 \right] \right\}}{\log (1.5)} \\ &\geq \frac{1}{2} \frac{\log (24)}{\log (1.5)} \geq 3.919 \approx 4 \end{aligned}$$

EXAMPLE 8.24 Determine the order and the poles of a low-pass Butterworth filter that has a -3dB bandwidth of 500 Hz and an attenuation of 40 dB at 1000 Hz.

Solution: Given

Passband edge frequency $f_1 = 500$ Hz, $\therefore \omega_1 = 2\pi f_1 = 1000\pi$

Gain at passband edge $k_1 = -3$ dB, $\therefore A_1 = 10^{k_1/20} = 10^{-3/20} = 0.707$

Stopband edge frequency $f_2 = 1000$ Hz, $\therefore \omega_2 = 2\pi f_2 = 2000\pi$

Gain at stopband edge $k_2 = -40$ dB, $\therefore A_2 = 10^{k_2/20} = 10^{-40/20} = 0.01$

Let the sampling frequency $f_s = 2000$ Hz.

The normalized frequencies are:

$$\omega_1 = 2\pi \frac{f_1}{f_s} = 2\pi \frac{500}{2000} = 0.5\pi$$

$$\omega_2 = 2\pi \frac{f_2}{f_s} = 2\pi \frac{1000}{2000} = \pi$$

For impulse invariant transformation,

$$\frac{\Omega_2}{\Omega_1} = \frac{\omega_2}{\omega_1} = 2$$

Therefore, order of the filter is:

$$N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] / \left[\frac{1}{A_1^2} - 1 \right] \right\}}{\log \left(\frac{\Omega_2}{\Omega_1} \right)}$$

$$\geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{(0.01)^2} - 1 \right] / \left[\frac{1}{(0.707)^2} - 1 \right] \right\}}{\log 2}$$

$$\geq \frac{1}{2} \frac{\log \{999/1\}}{\log(2)} \geq 6.64 \approx 7$$

The pole positions are:

$$s_k = \Omega_c e^{j \left[\frac{\pi}{2} + (2k+1)\pi/2N \right]}$$

$$= 1000 \pi e^{j \left[\frac{\pi}{2} + (2k+1)\pi/14 \right]}, \quad k = 0, 1, 2, 3, 4, 5, 6$$

where Ω_c is 3 dB cutoff frequency.

EXAMPLE 8.25 Determine the order of a Butterworth low-pass filter satisfying the following specifications:

$$f_p = 0.10 \text{ Hz}, \quad \alpha_p = 0.5 \text{ dB}$$

$$f_s = 0.15 \text{ Hz}, \quad \alpha_s = 15 \text{ dB}; f = 1 \text{ Hz}$$

Solution: Given

$$f_p = 0.10 \text{ Hz}, \quad \therefore \omega_p = \omega_1 = 2\pi f_p = 2\pi(0.1) = 0.2 \pi$$

$$f_s = 0.15 \text{ Hz}, \quad \therefore \omega_s = \omega_2 = 2\pi f_s = 2\pi(0.15) = 0.30 \pi$$

$$\alpha_p = \alpha_1 = 0.5 \text{ dB}, \quad \therefore k_1 = -0.5 \text{ dB}, \text{ so } A_1 = 10^{k_1/20} = 10^{-0.5/20} = 0.944$$

$$\alpha_s = \alpha_2 = 15 \text{ dB}, \quad \therefore k_2 = -15 \text{ dB}, \text{ so } A_2 = 10^{k_2/20} = 10^{-15/20} = 0.177$$

$$f = 1 \text{ Hz}, \quad \therefore T = \frac{1}{f} = \frac{1}{1} = 1 \text{ s.}$$

1. The type of transformation is not specified. Let us use bilinear transformation.

$$2. \quad \frac{\Omega_2}{\Omega_1} = \frac{\frac{2}{T} \tan \frac{\omega_2}{2}}{\frac{2}{T} \tan \frac{\omega_1}{2}} = \frac{\frac{1}{1} \tan \frac{0.3\pi}{2}}{\frac{1}{1} \tan \frac{0.2\pi}{2}} = \frac{1.019}{0.649} = 1.57$$

$$3. \quad N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] / \left[\frac{1}{A_1^2} - 1 \right] \right\}}{\log \left(\frac{\Omega_2}{\Omega_1} \right)}$$

$$\geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{0.177^2} - 1 \right] / \left[\frac{1}{0.944^2} - 1 \right] \right\}}{\log(1.57)}$$

$$\geq 6.16 \approx 7$$

so the order of the low-pass Butterworth filter is $N = 7$.